

1. Abstract

This report presents a systematic experimental study on audio-based gunshot classification using deep learning. The goal was to identify which firearm produced a given gunshot audio clip across 8 weapon classes. Three progressively refined approaches were explored: a baseline custom CNN, an improved CNN with regularisation and class balancing, and a transfer learning comparison against EfficientNet-B0, MobileNetV2, DeepGunshotCNN, and ViT-B/16. The best result — **81.40% test accuracy** — was achieved by the improved custom CNN, outperforming all transfer learning models. This finding demonstrates that domain-specific architectures trained from scratch on audio features can outperform general-purpose pretrained vision models when applied to audio spectrograms.

2. Introduction

Automated gunshot detection and weapon identification has significant applications in law enforcement, smart city surveillance, and security systems. Unlike image-based weapon detection, audio-based classification operates passively and can function in low-visibility conditions.

This experiment addresses the problem of **multi-class weapon identification** — given a short audio clip of a gunshot, classify which of 8 firearms fired it. The dataset contains 851 labelled WAV recordings across 9 weapon folders, with M16 and M4 merged into a single M-Family class due to acoustic similarity.

The core research question: *Does transfer learning from ImageNet-pretrained vision models improve performance on audio MFCC spectrograms, or does a custom CNN trained from scratch perform better?*

3. Dataset & Preprocessing

3.1 Dataset

Weapon Class	Samples	Label
AK-12	98	0
AK-47	72	1
IMI Desert Eagle	100	2
M16 + M4 (M-Family)	200	3
M249	99	4
MG-42	100	5
MP5	100	6
Zastava M92	82	7
Total	851	—

3.2 Audio Preprocessing Pipeline

All audio files were processed through a standardised pipeline before feature extraction:

Mono Conversion: Stereo files averaged across channels to produce single-channel audio.

Resampling: All files resampled to 48,000 Hz for consistency.

Duration Standardisation: Clips cropped to 2 seconds or zero-padded to fill 2 seconds (96,000 samples).

Pre-Emphasis: High-frequency boost applied (coefficient 0.97) to enhance transient characteristics of gunshots.

Amplitude Normalisation: Waveform rescaled to [-1.0, 1.0] range.

MFCC Extraction: 40 Mel Frequency Cepstral Coefficients computed (n_fft=1024, hop_length=512, n_mels=64), producing [1, 40, 188] tensors per clip.

4. Experiments

4.1 Experiment 1 — Baseline CNN

A custom 2-layer CNN was trained as a baseline. The architecture consisted of two convolutional blocks with BatchNorm and MaxPool, followed by fully connected layers with Dropout(0.3). The model was trained with CrossEntropy loss, Adam optimiser, and a ReduceLROnPlateau scheduler on a 75/25 train/val split.

4.2 Experiment 2 — Improved CNN (Best Model)

The improved CNN addressed three weaknesses identified in the baseline:

- Weighted CrossEntropy loss to handle class imbalance (AK-47: 72 samples vs M-Family: 200 samples)
- Cosine Annealing LR scheduler for smoother convergence
- Early stopping with patience=5 to prevent overfitting
- Feature-space augmentation on training set (random time shift + Gaussian noise)
- Wider FC layer (256 neurons vs 128) for greater representational capacity
- 70/15/15 train/val/test split for unbiased final evaluation

4.3 Experiment 3 — Transfer Learning Comparison

Four additional architectures were benchmarked against the improved CNN using identical data splits and training configuration:

EfficientNet-B0: ImageNet-pretrained, first conv layer adapted for single-channel MFCC input, classifier replaced with 8-class head.

MobileNetV2: Lightweight ImageNet model adapted for audio spectrograms.

DeepGunshotCNN: Deeper custom CNN with 3 conv blocks and adaptive pooling.

ViT-B/16: Vision Transformer pretrained on ImageNet-21k, linear probe fine-tuning only.

5. Results

5.1 Model Comparison

Model	Train Acc	Val Acc	Test Acc	Rank
Improved CNN ★	84.37%	80.31%	81.40%	#1
GunshotCNN (Baseline)	76.30%	75.59%	77.52%	#2
DeepGunshotCNN	78.15%	59.06%	73.64%	#3
EfficientNet-B0	76.30%	74.80%	73.64%	#3
MobileNetV2	73.45%	57.48%	70.54%	#5
ViT-B/16	54.29%	50.39%	54.26%	#6

★ Best performing model

5.2 Best Model — Per-Class Performance

Classification report on the held-out test set (Improved CNN, 81.40% accuracy):

Class	Precision	Recall	F1-Score	Support
AK-12	1.00	1.00	1.00	19
AK-47	0.89	0.80	0.84	10
IMI Desert Eagle	0.69	0.69	0.69	13
M-Family	0.79	0.77	0.78	35
M249	0.75	0.82	0.78	11
MG-42	0.92	0.86	0.89	14
MP5	0.58	0.65	0.61	17
Zastava M92	1.00	1.00	1.00	10
Macro Avg	0.83	0.82	0.82	129
Weighted Avg	0.82	0.81	0.82	129

6. Analysis & Key Findings

Custom CNN outperforms transfer learning

The Improved CNN (81.40%) beat all five competing models including EfficientNet-B0 (73.64%) and ViT-B/16 (54.26%). ImageNet-pretrained models do not transfer well to MFCC spectrograms because MFCCs have fundamentally different statistical properties to natural images.

ViT-B/16 performed worst

With only 54.26% test accuracy, ViT-B/16 struggled most. Vision Transformers require large datasets to learn useful attention patterns. With only ~600 training samples, the transformer failed to generalise.

DeepGunshotCNN overfit

DeepGunshotCNN had the highest train accuracy (78.15%) but dropped to 59.06% on validation — a 19% gap indicating severe overfitting. More parameters without sufficient data hurts performance.

Class imbalance matters

AK-47 (72 samples) and Zastava M92 (82 samples) showed divergent results — AK-47 had lower F1 (0.84) while Zastava M92 achieved perfect F1 (1.00). Weighted loss partially mitigated this but class imbalance remains the main limitation.

MP5 is the hardest class

MP5 consistently showed the lowest F1 across all models (0.61 in best model). MP5 is a submachine gun whose acoustic profile overlaps with several other weapons.

Perfect scores on AK-12 and Zastava M92

Both achieved 1.00 precision, recall, and F1 — indicating highly distinctive acoustic signatures that all models could reliably identify.

7. Conclusion

This study systematically compared 5 deep learning architectures for audio-based gunshot weapon classification. The key finding is that a well-regularised custom CNN trained from scratch on MFCC features achieves superior performance (81.40%) compared to transfer learning from ImageNet-pretrained models.

The improvements in Experiment 2 — weighted loss, cosine annealing, early stopping, and feature augmentation — contributed meaningfully to the 4% accuracy gain over the baseline. These techniques are directly applicable to any small-dataset audio classification problem.

Future work should focus on: (1) expanding the dataset to 500+ samples per class, (2) exploring audio-specific pretrained models such as PANNs or YAMNet, and (3) testing on real-world recordings with background noise to assess robustness.

8. Technical Stack

Component	Technology
Framework	PyTorch 2.x
Audio Loading	torchaudio / soundfile
Features	MFCC (40 coefficients, 48kHz)
Transfer Models	torchvision (EfficientNet-B0, MobileNetV2, ViT-B/16)
Evaluation	scikit-learn (classification_report, confusion_matrix)
Environment	Python 3.10, CUDA GPU